

650-291-4624
hirotaso92602@gmail.com

So Hirota

GitHub: soh09
LinkedIn: So Hirota
Website (soh09.github.io)

Education

Stanford University

M.S. Computer Science

September 2025 - June 2027

Stanford, CA

University of California, San Diego

B.S. in Data Science, Minor in Cognitive Science (GPA: 3.95)

September 2021 - June 2025

La Jolla, CA

Work Experience

Machine Learning Intern

NASA-JPL (Jet Propulsion Lab)

July 2025 - Present

Pasadena, CA

- Developing scalable quality assurance tools for the ARIEL space telescope mission's EXCALIBUR data pipeline, automating detection of anomalies in various scientific data products
- Generated synthetic residual datasets by perturbing orbital parameters, enabling supervised training of model
- Training a convolutional neural networks to regress deviation from true orbital parameters (in sigma units) and identify data issues that manual inspection would miss.

Data Science Intern

Franklin Templeton, Strategic Ventures

June 2024 - September 2024

San Mateo, CA

- Developed a financial data pipeline from the ground up using DataBricks, Snowflake, and PowerBI, resulting in an 80% reduction in manual data processing time and enabling informed decision-making by stakeholders
- Leveraged SQL, Python, and Llama models with tailored prompts to parse emails, slide decks, spreadsheets, and text files
- Held frequent meetings with the users to understand their needs and prioritized ease of use for non-technical users
- Created extensive documentation of the system for future users and engineers

Machine Learning Engineer

Argon Robotics

October 2023 - Present

La Jolla, CA

- Rapidly prototyped and delivered a CNN for weld seam detection within 2 weeks to be used in a production environment
- Advised the founder on neural network architectures like autoencoders and U-Net and identified the optimal solution, based on requirements like core objectives, performance, and data availability

Data Science Intern

menu, Inc

July 2023 - September 2023

Tokyo, JP

- Menu is a major food delivery technology company, mainly competing with "Uber Eats" in Japan
- Constructed a regression model to predict daily online time per delivery crew by factoring in weather conditions, seasonality, and historical data
- Led the development of a time and order flow simulation to simulate order statuses, delivery driver locations, and order queue, which facilitated the testing of novel order-driver matching algorithms

Projects

Dinosaur Nugget Anomaly Detector (1st Place at DataHacks)

April 2024

- Prototyped and trained a convolutional autoencoder in PyTorch to detect anomalous shapes in dinosaur-shaped nuggets
- Collaborated with team members to collect data, load/transform data, and pitch our project to a panel of judges
- Winning project out of 30 teams at the UCSD-hosted annual "DataHacks" hackathon

2048AI - Evolving Neural Networks

December 2023 - Current

- Implemented neural networks and the Neur-Evolution of Augmenting Topologies (NEAT) algorithm from scratch
- Designing fitness functions that balance human intuition and objective metrics to facilitate effective evolution
- Profiled the NN and NEAT implementation to identify inefficiencies in code, resulting in a 20% speedup in performance

American Sign Language Detection Glove (1st Place at HardHack)

April 2023

- Developed a glove that could measure the position of each finger to detect the ASL alphabet in real-time
- Trained a K-Nearest Neighbor algorithm that could detect signs with 85% accuracy using Scikit-Learn

Research Experience

Research Assistant

LLM Reasoning Lab by Prof. Zhiting Hu

September 2024 - Current

La Jolla, CA

- Proposed methods to extend the Process Reward Modeling paradigm for LLM fine-tuning by incorporating natural language feedback instead of scalar feedback, and reviewed relevant literature to explore the feasibility and novelty
- Implemented a best-of-n sampling strategy for a browser-enabled LLM agent and benchmarked performance on Visual WebArena.
- Created presentations to share findings from relevant literature for group-wide brainstorming sessions